

Possible improvements for Mountain NER

1. Two-stage mountain extraction. A catalog matcher first finds possible mountain names through aliases, normalization, and fuzzy matching. BERT, SpanMarker, or an LLM then checks each candidate in its sentence and removes false positives. This can improve recall without running a large model on every sentence.

2. Synthetic dataset for all mountains. Generate several clean sentence variants for every unique name in the combined mountain catalog. Deduplicate them against Few-NERD and add them only to training, never to validation or test. Compare the existing 196 examples with the full-catalog version using entity F1.

3. Conditional diffusion transformer. A custom denoising transformer can receive mountain names and generate synthetic contexts or BIO spans. It is a research option for difficult spelling and context cases, not the default solution. Keep it only if it beats normal BERT fine-tuning and the simpler two-stage pipeline on validation and test.

Possible improvements for Sentinel-2 image matching

1. Semi-supervised matching and other backbones. Use high-confidence LoFTR and MAGSAC matches as pseudo-labels for unlabeled seasonal pairs. Train a Siamese encoder or local matcher with contrastive or InfoNCE loss, reprojection consistency, and hard negatives. Compare LoFTR, SuperPoint plus LightGlue, DISK plus LightGlue, ResNet, Swin, and DINOv2 on geographic tile splits only.

2. Trainable Siamese scene-pair classifier. This is the fourth repository approach after SIFT, ORB, and LoFTR, but it solves a different task: it predicts whether two tiles represent the same area rather than returning keypoint correspondences. Train it with BCEWithLogitsLoss for same-area classification, or contrastive loss for retrieval, and select the decision threshold only on validation data. It should be evaluated separately with ROC-AUC, PR-AUC, precision, recall, and F1 after collecting enough geographically disjoint tile groups.